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class Product:
    def __init__(self, name, price, stock):
        self.name = name
        self.price = price
        self.stock = stock

    def update_stock(self, quantity):
        self.stock += quantity

    def __str__(self):
        return f"Product: {self.name}, Price: ${self.price:.2f},  
Stock: {self.stock}"

def add_product(products):
    name = input("Enter product name: ")
    price = float(input("Enter product price: "))
    stock = int(input("Enter product stock: "))
    new_product = Product(name, price, stock)
    products[name] = new_product
    print("Product added successfully!")

def update_stock(products):
    name = input("Enter product name to update: ")
    if name in products:
        quantity = int(input("Enter quantity to add/remove: "))
        products[name].update_stock(quantity)
        print("Stock updated successfully!")
    else:
        print("Product not found!")

def view_product(products):
    name = input("Enter product name: ")
    if name in products:
        print(products[name])
    else:
        print("Product not found!")

def main():
    products = {}
    while True:
        print("\n1. Add a new product")
        print("2. Update stock of an existing product")
        print("3. View product details")
        print("4. Exit")

        choice = input("Enter your choice (1-4): ")

        if choice == '1':
            add_product(products)
        elif choice == '2':

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        update_stock(products)
    elif choice == '3':
        view_product(products)
    elif choice == '4':
        print("Exiting the system. Goodbye!")
        break
    else:
        print("Invalid choice. Please try again.")

if __name__ == "__main__":
    main()

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1. Add a new product
2. Update stock of an existing product
3. View product details
4. Exit

Enter your choice (1-4): 1
 Enter product name: LAPTOP
 Enter product price: 10000
 Enter product stock: 10

Product added successfully!

1. Add a new product
2. Update stock of an existing product
3. View product details
4. Exit

Enter your choice (1-4): 2
 Enter product name to update: LAPTOP
 Enter quantity to add/remove: 2

Stock updated successfully!

1. Add a new product
2. Update stock of an existing product
3. View product details
4. Exit

Enter your choice (1-4): 3
 Enter product name: LAPTOP

Product: LAPTOP, Price: \$10000.00, Stock: 12

1. Add a new product
2. Update stock of an existing product
3. View product details
4. Exit

Enter your choice (1-4): 4

Exiting the system. Goodbye!